Constant Auxiliary Losses Considered Harmful: A Cautionary Tale in Belief-Learning Multi-Agent Reinforcement Learning

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Abstract

Attention-based belief encoders trained with constant-weight auxiliary prediction losses are a widespread design pattern in cooperative multi-agent reinforcement learning (MARL). We identify a directional gradient-interference pathology in this pattern under co-learning non-stationarity. In our 10M-step reruns of the published AERIAL architecture [31] on Overcooked Asymmetric Advantages, the 5-seed results exhibit a late-training drop: 4 of 5 seeds reach peak $\sim$482 and then drift down to Final50 $\sim$463 (drop 15.5 ± 3.3), while a fifth seed fails to converge at all (all-seed aggregate: drop 12.6 ± 6.6). To isolate the mechanism we construct VABL as a minimal architecture in the same class and run a 7-configuration ablation across three environment families (Overcooked, SMAX, MPE). The pathology manifests on Overcooked (Cohen's d up to 5.06) and SMAX ($d = 1.26$) but not on the simpler MPE (cooperative navigation), consistent with our theoretical prediction of environment-dependent instability thresholds. Three gradient-targeted interventions all sever the auxiliary-to-encoder gradient pathway and all converge to the no-aux regime, isolating that pathway as the necessary conduit for the instability. A gradient decomposition on VABL reveals the auxiliary gradient is small in magnitude ($< 25\%$ of the policy gradient) but oscillates erratically in direction, pointing to directional rather than magnitude-driven interference. A 60-run architecture sweep reproduces the effect at positive Cohen's d in two of four attention-based actor configurations (GRU + additive, LSTM + cross-attention), with the other two too high-variance at $n = 5$ to resolve an aux effect; mean pooling is immune and critic-side auxiliary placement reverses the sign. We additionally show that "coordination collapse" on Overcooked is largely a training-budget artifact (collapse $< 5\%$ at 10M steps). For practitioners, in the configurations we tested, any one of three gradient-pathway interventions is sufficient: $\lambda \leq 0.01$ or annealing, stop-gradient on the belief into the auxiliary head, or placing the auxiliary head on the centralized critic. Code, training logs, and per-seed JSON results for all 350+ runs are available at `https://anonymous.4open.science/r/aux-loss-considered-harmful`.

1 Introduction

Cooperative MARL on Overcooked is widely reported to collapse after reaching strong peak reward during training: MAPPO by 76% [44], AERIAL by 100% [31], with communication-based methods showing similar instability [10, 38]. These reports have motivated auxiliary prediction losses [20], attention-based belief encoders [31], and explicit communication channels as stabilization mechanisms. The resulting design pattern, a shared encoder trained jointly with a policy loss and

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a constant-weight auxiliary prediction loss, is now widespread beyond MARL, appearing in world models [15], multi-task RL [20], and self-supervised representation learning [37]. We find most of the reported Overcooked collapse to be a training-budget artifact. In our sample-efficiency measurements (four VABL ablation configurations, 5 seeds each, 80 runs), collapse drops from $\sim$46% at 500K environment steps to $<$6% at 2.5M and $<$3% at 10M, and MAPPO, TarMAC, and CommNet all converge stably (collapse $\leq 3\%$) by 10M.

One architectural class still degrades at 10M steps. In our 10M-step reruns of AERIAL [31], the 5-seed results exhibit a striking late-training drop: four seeds reach a peak of $\sim$482 and then drift down to Final50 $\sim$463 (drop 15.5 ± 3.3), while a fifth seed never converges, plateauing at $\sim$360 from roughly 25% of training onward (5-seed aggregate drop 12.6 ± 6.6). This 10-20 point residual drop is a distinct, gradient-mediated pathology of the architectural class, not the short-horizon collapse reported in earlier literature.

Isolating the mechanism. To investigate this pathology under controlled ablation, we construct **VABL** (Variational Attention-based Belief Learning), a minimal architecture in the same class. VABL is deliberately simpler than AERIAL: it has no learned message channel, no hidden-state sharing, and a cleanly separable auxiliary pathway, which lets us toggle the auxiliary loss and intervene on its gradient flow without architectural changes. In our 7-configuration VABL ablation on Overcooked AA at 10M steps, the attention + constant-aux configuration is the only one that exhibits late-training instability. Best reward ($\sim$484) matches all alternatives, but Final50 drops to 463 ± 8.6 with 2–5$\times$ higher cross-seed variance. The pathology reproduces on a second Overcooked layout (Cohen's $d = 5.06$ on Cramped Room) and on SMAX ($d = 1.26$) but not on MPE simple_spread, consistent with our theoretical prediction that the instability threshold depends on the complexity of co-learning dynamics (Proposition 1).

Three gradient-targeted interventions, **(i)** λ-annealing, **(ii)** stop-gradient on the belief input to the auxiliary head, and **(iii)** both combined, converge to a single recovered regime (470–474), statistically equivalent to removing the auxiliary loss. This convergent evidence causally implicates the auxiliary gradient flowing into the attention-based encoder. A CIFAR-100 cross-domain test finds the pathology absent in supervised learning, narrowing it to MARL's co-learning non-stationarity.

Our contributions: **(1)** identification of a directional gradient-interference pathology in the attention + constant-auxiliary design pattern under MARL co-learning, first observed in the published AERIAL architecture [31] and then isolated under controlled ablation in a minimal test bed (VABL), including environment-dependence (present on Overcooked and SMAX, absent on MPE) and λ sensitivity between λ=0.01 (stable) and λ=0.05 (unstable); **(2)** a theoretical analysis (Proposition 1) whose key bound scales with $\mathrm{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$, paired with a gradient decomposition showing the auxiliary gradient is small in magnitude ($< 25\%$) but oscillates erratically in direction; **(3)** three gradient-targeted fix paths that all sever the auxiliary-to-encoder gradient pathway and all recover to the no-aux regime, yielding concrete design rules ($\lambda \leq 0.01$ *or anneal; stop-gradient on the belief into the auxiliary head; or place the auxiliary head on the centralized critic instead of the actor*), generalized in a 60-run architecture sweep (Table 11) across recurrence and attention variants; **(4)** evidence that the much-reported Overcooked "coordination collapse" is largely a training-budget artifact, plus a reproducible 350+-run, 5-seed benchmark across three MARL environment families with anonymized code, training logs, and per-run result files at `https://anonymous.4open.science/r/aux-loss-considered-harmful`.

2 Related Work

Auxiliary losses in RL. UNREAL [20] pioneered auxiliary tasks for representation shaping; subsequent work uses world-model losses [15], self-predictive representations [37], and contrastive objectives [23]. Lyle et al. [28] provide a theoretical analysis of how auxiliary tasks shape representation dynamics in TD-based deep RL, and Du et al. [12] and Lin et al. [24] both argue that constant auxiliary weights are suboptimal and propose adaptive schemes (cosine-similarity gating and online weighting, respectively) as remedies. In the MARL setting, XP-MARL [43] frames an auxiliary prioritization signal explicitly as a tool to manage co-learning non-stationarity. We instead *characterize* the pathology: under MARL co-learning, constant weights destabilize attention-based encoders at long horizons, and three distinct gradient interventions converge to the same recovered regime.

Attention in MARL. AERIAL [31] uses attention over teammate hidden states; MAAC [19] places attention in the centralized critic; MAT [42] uses an encoder-decoder for centralized policy learning. None pair attention-based *decentralized* belief encoding with an explicit auxiliary loss, the specific combination our ablation targets.

Stop-gradient in SSL. BYOL [14], SimSiam [6], and DINO [4] use stop-gradient to prevent representation collapse. Our stop-gradient fix path is analogous: it prevents the auxiliary loss from destabilizing the belief encoder via gradient interference.

Gradient interference. PCGrad [45] projects conflicting task gradients; GradNorm [7] balances task weights dynamically; HyperMARL [39] introduces agent-conditioned hypernetworks to decouple cross-agent gradient flow in the actor parameters. These methods address *cross-task* or *cross-agent* interference: two losses (or two agents) at the same time step pulling in opposite directions. Multi-timescale learning [29] addresses MARL non-stationarity differently, by slowing partner updates so that each agent sees a quasi-stationary target.

None of these techniques directly targets the mechanism we identify. PCGrad and GradNorm operate on the *magnitudes* of concurrent task gradients, but the auxiliary gradient in our pathology is already small (magnitude ratio $< 25\%$); the harm is in its *direction*, which oscillates from step to step under co-learning. HyperMARL decouples gradients *across agents*, but our pathology is intra-agent: the auxiliary and policy objectives share the same actor's belief encoder. Multi-timescale learning reduces the rate at which partner policies drift but does not remove the aux-to-encoder gradient pathway, which is what our three fix paths intervene on. Our finding is qualitatively different from all of these: the interference is *temporal* (the auxiliary gradient is benign early but becomes destabilizing late in training under co-learning non-stationarity), *directional* rather than magnitude-driven (the auxiliary gradient is small but erratically oriented), and *environment-dependent* (present on complex coordination tasks, absent on simpler ones). These distinctions are not captured by existing gradient-surgery frameworks.

Coordination collapse. Overcooked [5] is widely used for evaluating cooperative MARL [18], with prior reports of severe degradation for MAPPO [44], value decomposition [32], and communication methods [10]. Cooperative MARL has long been known to suffer named coordination pathologies, most prominently relative overgeneralization [41], where partners' random exploration biases joint-action value estimates toward suboptima. Deep RL more broadly exhibits late-training degradation through mechanisms like capacity loss [27]. Our 10M-step evaluation reveals that most reported MARL "collapse" results reflect insufficient training; the residual instability is a distinct, gradient-mediated pathology.

3 Background: Attention-Based Belief Learning

The design pattern we investigate, attention-based belief encoders trained with constant-weight auxiliary prediction losses, appears in multiple published methods including AERIAL (attention over teammate hidden states, auxiliary recurrence loss) [31] and is structurally compatible with the broader family of world-model and self-predictive representation architectures that use constant auxiliary weights [15, 37, 20]. We construct **VABL** (Variational Attention-based Belief Learning) as a minimal architecture in this class, operating on cooperative Dec-POMDPs [30], and chosen to be as small and cleanly ablatable as possible so that attention, the auxiliary loss, and the gradient flow between them can be manipulated independently.

Belief update. Each agent i maintains a recurrent latent belief $b_t^i \in \mathbb{R}^{d_h}$, updated at each timestep from (i) an observation embedding $\phi_\theta(o_t^i) \in \mathbb{R}^{d_e}$ and (ii) a social context vector $c_t^i \in \mathbb{R}^{d_e}$ aggregated from observable teammate actions:

$$b_t^i = \mathrm{GRU}\left([\phi_\theta(o_t^i) \| c_t^i],\ b_{t-1}^i\right), \tag{1}$$

where GRU denotes a gated recurrent unit [8].

Attention context. The social context is computed via multi-head attention [40] over action embeddings of visible teammates:

$$c_t^i = \mathrm{MHA}\left(\text{query} = W_q b_{t-1}^i, \quad \text{keys/values} = \{\psi_\theta(a_{t-1}^j) + e_j\}_{j \neq i}\right), \tag{2}$$

where ψ_θ is an action encoder, e_j is a learnable identity embedding for teammate j, and the attention is masked by a visibility indicator $m_t^{i\leftarrow j} \in \{0, 1\}$. When no attention is used (the mean-pooling ablation), c_t^i is the unweighted average of visible teammate action embeddings.

Policy and auxiliary heads. The policy $\pi_\theta(a_t^i \mid b_t^i)$ is a linear head on the belief. The auxiliary predictor $\hat{\pi}_\varphi(\cdot \mid b_t^i)$ is a 2-layer MLP that predicts each visible teammate's next action from the current belief, trained with cross-entropy:

$$\mathcal{L}_{\text{aux}}^i = \mathbb{E}\left[\sum_{j\neq i} m_{t+1}^{i\leftarrow j}\, \text{CE}(\hat{\pi}_\varphi(\cdot \mid b_t^i),\, a_{t+1}^j)\right]. \tag{3}$$

Training objective. The total loss combines a PPO policy gradient [36], a centralized value loss, entropy regularization, and the auxiliary loss:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + c_v\mathcal{L}_{\text{value}} + c_e\mathcal{L}_{\text{entropy}} + \lambda\sum_i \mathcal{L}_{\text{aux}}^i, \tag{4}$$

where $\lambda = 0.05$ is the auxiliary loss weight. In the standard configuration, λ is held constant throughout training.

Gradient pathway. The auxiliary loss gradient flows through the belief b_t^i and into the attention mechanism, the GRU, and the observation encoder ϕ_θ. This is the gradient pathway that our ablation (Section 6) identifies as the source of the late-training instability.

4 Theoretical Analysis: Directional Interference

We develop a minimal analytical model that predicts the empirical findings. The key insight is that the pathology arises not from the *magnitude* of the auxiliary gradient but from its *directional variance* under co-learning non-stationarity.

Gradient decomposition near convergence. Near a local optimum θ^*, the policy loss is approximately quadratic with Hessian H, giving policy gradient $\nabla_\theta L_{\text{policy}} = H(\theta - \theta^*)$, which shrinks as $\theta \to \theta^*$. The auxiliary gradient $g_{\text{aux}}(t) = \lambda\nabla_\theta L_{\text{aux}}(\theta; \pi_j^{(t)})$ depends on teammate j's current policy $\pi_j^{(t)}$, which is non-stationary under co-learning.

Noise decomposition. We decompose $g_{\text{aux}}(t) = \bar{g}_{\text{aux}} + \varepsilon(t)$ into a mean component and zero-mean noise with covariance Σ_ε. The noise $\varepsilon(t)$ arises specifically from co-learning non-stationarity [16, 29]: if teammate j's policy were frozen, $\varepsilon \to 0$. This non-stationarity is a defining feature of cooperative MARL and more broadly of any co-learning multi-agent system. Prior work has tackled it through multi-timescale learning rather than directly through a gradient-level analysis of how auxiliary objectives propagate into shared encoders.

Stochastic dynamics. Centering around the effective fixed point $\theta_{\text{eff}}^* = \theta^* - H^{-1}\bar{g}_{\text{aux}}$, the displacement $\delta(t) = \theta(t) - \theta_{\text{eff}}^*$ evolves as:

$$\delta(t{+}1) = (I - \alpha H)\,\delta(t) - \alpha\,\varepsilon(t), \tag{5}$$

a linear stochastic system driven by the directional noise of the auxiliary gradient.

Intuitively, given an auxiliary gradient with directional noise Σ_ε and a policy loss with minimum curvature $\lambda_{\min}(H)$, the bound below says the late-training parameter variance grows with the noise (numerator) and shrinks with the curvature in the flattest direction (denominator): a flatter landscape amplifies the noise. The proposition is *descriptive*, not a first-principles prediction: it relates two measurable quantities under a simplifying assumption.

Proposition 1 (Steady-state variance; descriptive linear model)**.** *Consider the linearized dynamics of Eq.* (5) *with noise $\varepsilon(t)$ (Cov $= \Sigma_\varepsilon$, treated as i.i.d. for tractability) and spectral radius $\rho(I - \alpha H) < 1$. Given these primitives, the steady-state parameter variance satisfies:*

$$\mathbb{E}[\|\delta\|^2] \;\propto\; \frac{\alpha^2\,\text{tr}(\Sigma_\varepsilon)}{\lambda_{\min}(H)}. \tag{6}$$

Assumptions and scope. This is a descriptive model, not a first-principles prediction of the pathology. We do not derive Σ_ε from a concrete teammate-drift generative process; we treat it as an empirical primitive measured in our gradient decomposition (Figure 1). The quadratic loss and i.i.d. noise assumptions are simplifications; real PPO dynamics are nonlinear and the noise is correlated. We establish two things about this model. First, a controlled synthetic experiment that directly constructs a linear stochastic system with prescribed H and Σ_ε recovers the variance bound exactly (Appendix D, Figure 6), confirming the algebra is correct. Second, the qualitative relationships implied by Eq. (6), namely that increasing Σ_ε or decreasing $\lambda_{\min}(H)$ worsens stability, and that driving $\Sigma_\varepsilon \to 0$ recovers stability, are consistent with the seven experimental observations enumerated below. The model is useful because its qualitative implications match the data, and because it explains why the fix paths work: each severs the aux-to-encoder gradient pathway and therefore drives the effective Σ_ε toward zero in the subspace that couples to the encoder.

Proof sketch. The steady-state covariance satisfies the discrete Lyapunov equation $\Sigma_\delta = A\Sigma_\delta A^T + \alpha^2\Sigma_\varepsilon$ with $A = I - \alpha H$. Diagonalizing yields $\mathrm{tr}(\Sigma_\delta) = \alpha^2 \sum_i \sigma^2_{\varepsilon,i}/(2\alpha h_i - \alpha^2 h_i^2)$, dominated by $\lambda_{\min}(H)$. Full derivation in Appendix A. □

Qualitative implications and empirical consistency. We list each qualitative implication of Eq. (6) and the corresponding experimental observation. We emphasize that these are consistencies, not *a priori* predictions: the observations identify directions in which the data agrees with the descriptive model, and give us reasons to take the bound's structure seriously even though its assumptions are simplifications.

1. **Implication:** Attention amplifies Σ_ε (multiplicative belief-query coupling) $\Rightarrow$ Full VABL should have the highest variance. **Consistent with:** Table 2, std 8.55 vs. 1.86–3.72 for alternatives.
2. **Implication:** $\lambda \to 0$ sends $\Sigma_\varepsilon \to 0 \Rightarrow$ annealing should recover stability. **Consistent with:** B_anneal Final50 $473.66 \approx$ A_no_aux 473.45 (Table 2).
3. **Implication:** Stop-gradient sets $\Sigma_\varepsilon = 0$ exactly in the encoder subspace $\Rightarrow$ should also recover. **Consistent with:** B_stopgrad Final50 471.60, lowest std (2.13).
4. **Implication:** Fixed auxiliary targets (supervised) $\Rightarrow \Sigma_\varepsilon \to 0$ naturally $\Rightarrow$ no pathology. **Consistent with:** CIFAR-100 shows < 0.5 point aux effect (Appendix B).
5. **Implication:** Complex co-learning $\Rightarrow$ large $\Sigma_\varepsilon \Rightarrow$ above threshold; simpler $\Rightarrow$ below. **Consistent with:** pathology present on Overcooked and SMAX (complex coordination); absent on MPE (simple navigation).
6. **Implication:** Smaller $\lambda \Rightarrow$ smaller $\mathrm{tr}(\Sigma_\varepsilon) \Rightarrow$ below threshold. **Consistent with:** λ=0.01 has std 1.67 (stable); λ=0.05 has std 8.55 (unstable) at the two values we tested.
7. **Implication:** The mechanism is directional (Σ_ε), not magnitude ($\|g_{\mathrm{aux}}\|$). **Consistent with:** Figure 1 shows $\|g_{\mathrm{aux}}\|/\|g_{\mathrm{PPO}}\| < 0.25$ but $\cos(g_{\mathrm{PPO}}, g_{\mathrm{aux}})$ oscillates in $[-0.47, +0.33]$ (5 seeds).

Empirical gradient evidence. Figure 1 plots the gradient decomposition over training for the Full VABL configuration on Overcooked AA. The pattern is consistent across all 5 seeds: the cosine similarity between PPO and auxiliary gradients oscillates erratically in $[-0.47, +0.33]$ while the magnitude ratio stays below 0.25. The auxiliary gradient *decreases* over training as the predictor improves, yet the directional instability persists, confirming the mechanism is directional noise, not magnitude-driven.

5 Methodology

5.1 Ablation Matrix

We run the VABL test bed in a 2×2 ablation over the two components that define the architecture's distinctive design:

- **Attention** vs. **mean pooling** for the social context aggregation.

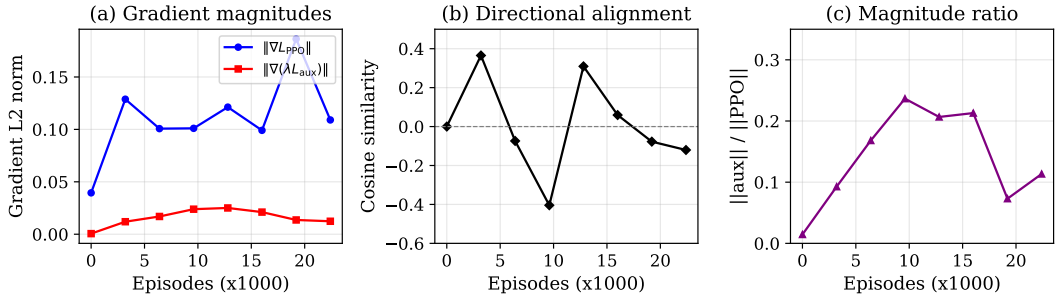

Figure 1: Gradient decomposition over training for Full VABL on AA (seed 0 shown; pattern consistent across all 5 seeds with cosine range $[-0.47, +0.33]$). **(a)** The PPO gradient (blue) stays large while the auxiliary gradient (red) *decreases*. **(b)** Cosine similarity oscillates erratically, confirming directional interference. **(c)** The magnitude ratio is always < 0.25; the pathology is not magnitude-driven.

- **Auxiliary loss on** ($\lambda = 0.05$, constant) vs. **auxiliary loss off** ($\lambda = 0$).

This yields four configurations: **Full** (attention + aux), **No Attention** (mean pool + aux), **No Aux** (attention only), and **Neither** (mean pool only).

5.2 Fix-Path Interventions

To test whether the observed pathology is gradient-mediated, we apply three interventions to the Full configuration that modify only the gradient flow, not the architecture or loss function:

- **Anneal**: linearly decay λ from 0.05 to 0 over the first 50% of training iterations. The auxiliary loss still fires during early training but phases out before late-training instability manifests.
- **Stop-gradient**: apply `stop_gradient` to the belief b_t^i before it enters the auxiliary head. The auxiliary loss still trains its own MLP weights but cannot propagate gradients into the belief encoder. This intervention affects *only* the gradient flow.
- **Anneal + stop-gradient**: both interventions combined.

5.3 Environments

We evaluate on three environment families to test generalization:

Overcooked [5]: two layouts, *Asymmetric Advantages* (asymmetric roles, 2 agents, horizon 400) and *Cramped Room* (spatial bottleneck). 25,024 episodes (≈10M env steps), 64 vectorized parallel environments via JAX `vmap`.

MPE simple_spread [26]: cooperative navigation with 3 symmetric agents (5 discrete actions, obs dim 18, horizon 25). 100,000 episodes (≈2.5M env steps).

SMAX 3v3 [33]: combat coordination with 3 allied agents against 3 heuristic enemies (obs dim 75, horizon 100). 50,000 episodes (≈5M env steps). SMAX is the JAX reimplementation introduced in JaxMARL, inspired by but distinct from the original SMAC [34]; it tests a fundamentally different coordination domain (tactical combat vs. cooking/navigation).

5.4 Baselines

We compare against four methods spanning three paradigms: **MAPPO** [44] (decentralized policy gradient), **TarMAC** [10] (targeted explicit communication), **CommNet** [38] (broadcast communication), and **AERIAL** [31] (attention over teammate hidden states). All baselines are reimplemented in JAX [3] using the JaxMARL framework [33] and trained with the same hyperparameters, vectorized environment wrapper, and evaluation protocol.

5.5 Cross-Domain Test

To test whether the pathology generalizes beyond MARL, we replicate the 2×2 ablation on CIFAR-100 [22] image classification with a small Vision Transformer [11] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head (4-way rotation prediction) [13]. The auxiliary loss uses the same $\lambda = 0.05$ constant weight.

5.6 Metrics

- **Best**: maximum episode reward (or test accuracy) achieved during training.
- **Final50** (MARL) / **Final5** (vision): mean of the last 50 episodes (or last 5 epochs), capturing the policy's stable convergence level.
- **Collapse %**: $(\text{Best} - \text{Final})/\text{Best} \times 100$.
- We report mean $\pm$ population standard deviation over 5 random seeds for all MARL experiments, and Cohen's d [9] for effect sizes.

6 Results

6.1 Empirical Observation in AERIAL

We first document the phenomenon in the published AERIAL [31] architecture on Overcooked Asymmetric Advantages at 10M training steps, 5 seeds. Table 1 reports the per-seed peaks and Final50 values from our reruns.

Table 1: Per-seed AERIAL trajectories on Overcooked Asymmetric Advantages (10M steps). Four of five seeds show the characteristic pattern: reach a peak of $\sim$482, then drift down 12–21 points by end of training. The fifth seed fails to converge at all, plateauing at $\sim$360 from roughly 25% of training onward. The four healthy seeds average a 15.5-point late-training drop, larger than Full VABL's 11.6 (Table 2, Group A).

Seed	Peak	Peak position	Final50	Drop
0	476.6	59%	460.1	16.5
1	361.5	100%	360.3	1.2[‡]
2	481.4	79%	468.5	12.9
3	478.0	42%	457.4	20.6
4	478.2	65%	466.2	12.0
4 healthy seeds (mean $\pm$ std)	478.6 ± 1.7	–	463.0 ± 4.4	15.5 ± 3.3
All 5 seeds (mean $\pm$ std)	455.1 ± 46.9	–	442.5 ± 41.3	12.6 ± 6.6

[‡]Seed 1 never converged: smoothed reward plateaued at $\sim$330 by 25% of training and only slowly rose to 360 by end. The "drop" of 1.2 reflects the fact that the run had no real peak to fall from. We report both the 4-healthy-seed aggregate and the full 5-seed aggregate; the paper's quantitative claims about late-training instability refer to the 4 healthy seeds.

The late-training drop signature holds under both aggregates (15.5 ± 3.3 on four healthy seeds, 12.6 ± 6.6 on all five) and is comparable in magnitude to what Full VABL exhibits under controlled ablation (Section 6.2, 11.6 ± 7.2), consistent with a shared underlying mechanism. A controlled ablation of AERIAL's auxiliary pathway would require architectural changes that break the method, so to isolate the mechanism we turn to VABL, which shares AERIAL's critical features (attention-based belief encoder + constant auxiliary loss) while admitting clean ablation of each component.

6.2 Isolating the Pathology in VABL

Table 2 presents the 7-configuration ablation on Overcooked Asymmetric Advantages at 10M steps. The Full configuration (attention + constant auxiliary loss) is the only configuration that exhibits the pathology: its Final50 of 463.21 ± 8.55 is significantly lower than every alternative (Cohen's $d \geq 0.88$), and its cross-seed standard deviation is 2–5$\times$ larger. Critically, all seven configurations reach the same Best reward ceiling (482–485), confirming that the pathology is purely a late-training stability issue, not a peak-performance limitation.

Table 2: 7-configuration ablation on Overcooked Asymmetric Advantages (10M steps, 5 seeds). Group A: 2×2 ablation over attention and auxiliary loss. Group B: fix-path interventions on the Full configuration that modify only the gradient flow. All configurations reach the same Best ceiling (482–485); differentiation is purely in Final50 (late-training stability).

Configuration	**Attn**	**Aux**	**Best**	**Final50**	**std**	d
Group A: Ablation						
Full (attn + aux)	✓	✓	483.8	463.21	**8.55**	–
Neither (mean only)	–	–	482.6	469.70	3.72	0.88
No Attention (mean + aux)	–	✓	483.8	471.76	1.86	1.24
No Aux (attn only)	✓	–	484.4	**473.45**	3.62	**1.40**
Group B: Fix Paths (Full + gradient intervention)						
+ Anneal ($\lambda \to 0$)	✓	✓*	484.4	473.66	3.59	1.43
+ Stop-gradient	✓	✓†	482.6	471.60	**2.13**	1.20
+ Both	✓	✓*†	484.4	472.35	2.21	1.31

Cohen's d is vs. Full. *Aux λ linearly decayed $0.05 \to 0$ over first 50% of training. †`stop_gradient` on belief into aux head.

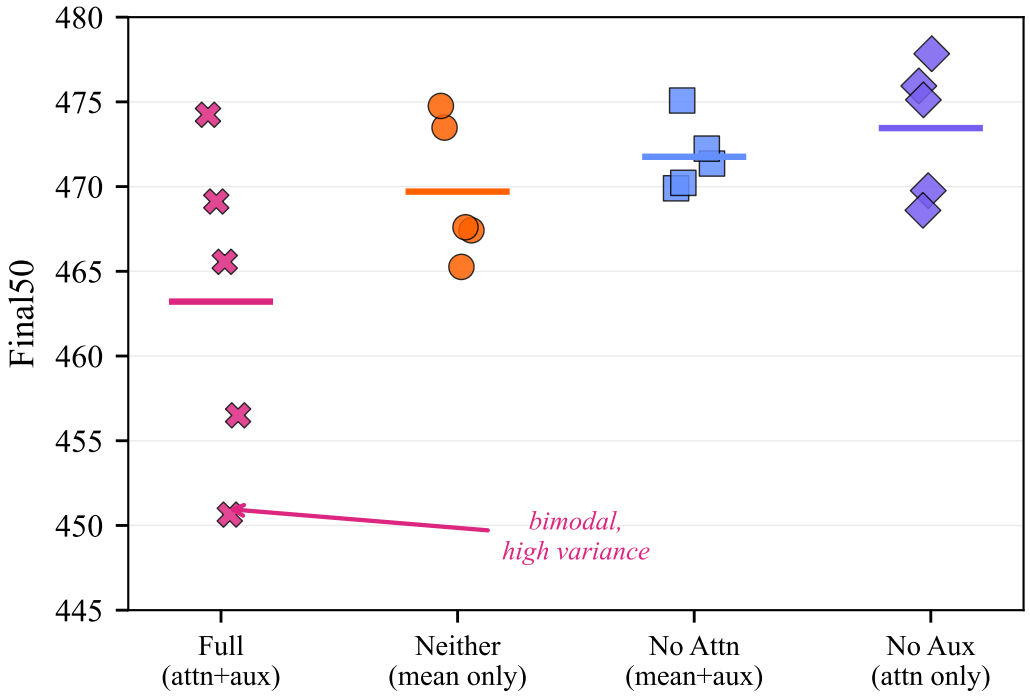

Figure 2: Per-seed Final50 on Overcooked AA (10M steps, 5 seeds). Full VABL (magenta) has a cross-seed std of 8.55 (2–5× larger than the other configurations) with individual seeds drifting 4–23 points below their own peaks; all other configurations cluster within a std of 3.7. Horizontal lines show means. Marker shapes vary for grayscale readability.

The 2×2 interaction is clean: neither attention alone (473.45 ± 3.62) nor the auxiliary loss alone (471.76 ± 1.86) causes the pathology. The instability requires *both* components; it is an interaction effect (full 2×2 in Appendix C).

6.3 Convergent Fix Paths

The three fix paths in Table 2 (Group B) all converge to Final50 in $[471.60, 473.66]$, a 2.06-point spread smaller than the seed envelope of any individual configuration, and statistically equivalent to removing the auxiliary loss entirely (A_no_aux: 473.45). All three interventions intervene on the

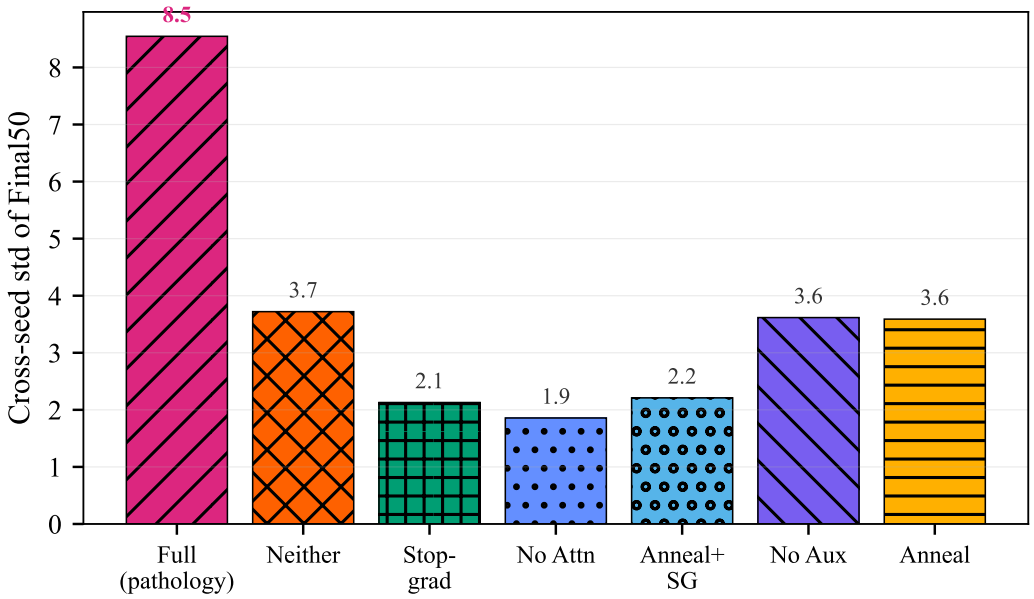

Figure 3: Cross-seed standard deviation of Final50 across all 7 configurations. Full VABL (magenta) has 8.55 std, 2–5$\times$ larger than any alternative. The variance pattern is monotonic in residual gradient flow: full flow (Full) $\rightarrow$ phased-out flow (Anneal) $\rightarrow$ no flow (Stop-grad, No Aux). Hatching patterns distinguish bars in grayscale.

Table 3: Cross-layout generalization on Overcooked Cramped Room (10M steps, 5 seeds). All methods saturate at Best $= 417$; differentiation is purely in Final50. The pathology reproduces with an even larger effect size (Cohen's $d = 5.06$).

Configuration	**Final50**	**std**	d **vs Full**	**95% CI on** d
Full (attn + aux)	409.70	1.41	–	–
No Attention (mean + aux)	411.05	2.57	$+0.58$	$[-0.75, +2.35]$
Neither (mean only)	416.02	1.33	$+4.13$	$[+2.43, +8.32]$
No Aux (attn only)	**416.18**	**0.79**	$\mathbf{+5.07}$	$[+3.04, +10.74]$

same pathway, severing auxiliary gradient flow into the attention-based belief encoder: annealing removes it over time, stop-gradient cuts it at its source, and the combined intervention does both. The convergence to the no-aux regime therefore identifies the aux-to-encoder gradient pathway as the *necessary conduit* for the pathology, rather than confirming a specific functional form for how interference is generated. Stop-gradient is the tightest of the three in that it modifies *only* the gradient flow (not the architecture, loss function, forward pass, or reward signal) and reduces the cross-seed standard deviation from 8.55 to 2.13, a $4\times$ reduction.

6.4 Cross-Layout Generalization

The pathology reproduces on a second Overcooked layout (Cramped Room) with an even larger effect size: Full VABL at 409.70 ± 1.41 vs. No Aux at 416.18 ± 0.79 (Cohen's $d = 5.07$, 95% parametric bootstrap CI $[+3.04, +10.74]$; see Appendix C). On Cramped Room, all methods saturate at the same Best reward of 417; the entire differentiation is in Final50 stability. The auxiliary loss is harmful regardless of whether attention is used, though the effect is strongest in the Full configuration. The No Attention comparison has a CI that crosses zero, matching the Asymmetric Advantages pattern where the non-attention cells are the harder ones to resolve at $n = 5$.

6.5 The Training-Budget Artifact

Table 4 presents the 6-method benchmark at 10M steps on both layouts. Two findings stand out:

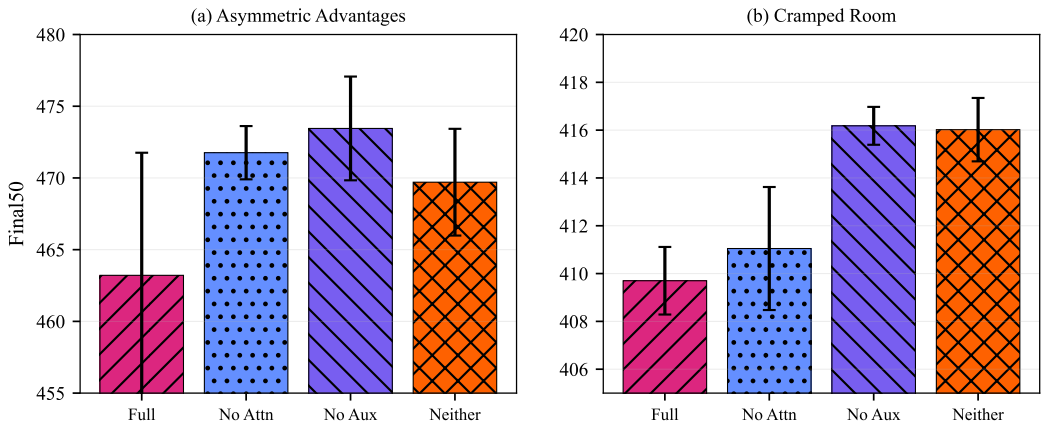

Figure 4: Cross-layout comparison of the 4-config ablation. The pathology (Full VABL, magenta) is the worst on both layouts. The effect is cleaner on Cramped Room where all methods saturate at Best $= 417$.

Table 4: 6-method benchmark at 10M steps (5 seeds each). VABL uses the best ablation config (No Aux = attention only). Communication-based methods (TarMAC, CommNet) dominate in mean reward but with $30\times$ higher cross-seed variance than VABL. On Cramped Room, all methods saturate at the same ceiling. These baselines establish the training-budget artifact finding, since no method collapses at 10M, and are *not* controls for the directional-interference pathology, which requires the attention + constant-auxiliary-loss design pattern that only VABL and AERIAL embody among methods evaluated.

		Asymm. Advantages		**Cramped Room**	
Method	**Comm?**	**Final50**	**std**	**Final50**	**std**
TarMAC	yes	**783.72**	156.10	414.73	2.78
CommNet	yes	773.24	148.17	416.30	0.43
MAPPO	no	548.23	153.09	**417.00**	**0.00**
VABL (No Aux)	no	473.45	**3.62**	416.18	0.79
AERIAL	no*	442.51	41.30	416.43	0.55
VABL (Full, pathology)	no	463.21	8.55	409.70	1.41

*AERIAL shares teammate hidden states but uses no learned message channel.

Communication-based methods dominate in mean but with enormous seed variance. TarMAC (783.72 ± 156.10) and CommNet (773.24 ± 148.17) achieve the highest Final50 on Asymmetric Advantages, but their cross-seed standard deviations exceed 148 points, roughly $30\times$ larger than VABL's. Single-seed comparisons between communication and communication-free methods are uninformative at this scale (given $\sigma > 148$).

VABL has the tightest cross-seed variance of any method. Even the pathological Full configuration ($\sigma = 8.55$) has far lower variance than MAPPO ($\sigma = 153.09$), TarMAC ($\sigma = 156.10$), CommNet ($\sigma = 148.17$), or AERIAL ($\sigma = 41.30$). The fixed VABL variants (std 1.86–3.72) are uniquely reproducible across seeds.

6.6 Architecture Generalization

A 60-run architecture sweep (Table 11 in Appendix F) swaps recurrence and attention independently on Overcooked AA. Three findings: **(i)** the pathology generalizes: LSTM + cross-attention ($d = 0.80$) and GRU + additive attention ($d = 1.01$) both reproduce it; **(ii)** mean pooling is immune ($d = -0.30$), isolating learned attention weights as the vulnerability; **(iii)** placing the auxiliary head on the centralized critic (MAAC-style [19]) *reverses* the effect ($d = -0.79$): the pathology is specific to auxiliary gradients flowing through the actor's shared belief encoder.

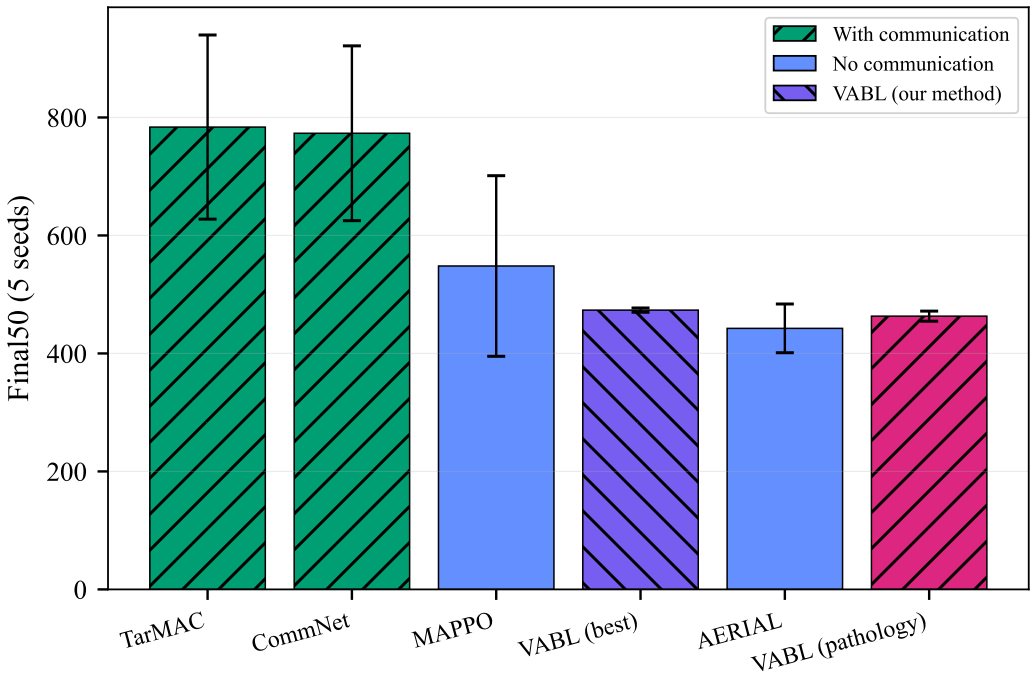

Figure 5: 6-method benchmark on Overcooked AA at 10M steps. Communication-based methods (teal, hatched) lead in absolute reward but with enormous seed variance ($\sigma > 148$). VABL (purple) has the tightest variance of any method ($\sigma < 4$).

Design recommendation. Taking the sweep together with the fix-path results, we offer the following recommendation for MARL practitioners: decentralized actor architectures that combine learned attention with a constant-weight auxiliary loss on the actor-side shared encoder may be at risk for the late-training directional-interference pathology we document, in the configurations we tested. Of the six sweep configurations, two (GRU + additive, LSTM + cross-attention) showed the effect at positive Cohen's d, two (GRU + self-attention, FF + cross-attention) had intrinsic cross-seed variance too high to resolve an aux effect, one (mean pooling) was immune, and one (critic-side aux) reversed the sign. Within the resolvable cases, the pattern is consistent: any of three interventions (mean pooling instead of learned attention, critic-side instead of actor-side auxiliary placement, or severing the aux-to-encoder gradient pathway via annealing or stop-gradient) suffices to recover the no-aux regime.

Environment-dependent threshold and λ sensitivity. We test on SMAX 3v3 (combat, 50K episodes) and MPE simple_spread (navigation, 100K episodes). On SMAX, the pathology **reproduces**: No Aux (11.15 ± 0.41) beats Full (10.60 ± 0.46), Cohen's $d = 1.26$, matching Overcooked's direction. On MPE, the pathology is **absent**: Full VABL (-18.82 ± 0.99) performs comparably to No Attn (-18.59 ± 1.08), with no penalty for the attention + aux combination. This pattern, present on Overcooked and SMAX, absent on MPE and in supervised learning (Appendix B), matches Proposition 1's environment-dependent threshold. We interpret the contrast mechanistically: Overcooked AA and SMAX 3v3 both require precise temporal synchronization of asymmetric roles, so teammate j's policy $\pi_j^{(t)}$ changes rapidly whenever either agent restructures its behavior, amplifying the non-stationarity that drives Σ_ε. MPE simple_spread has symmetric, distance-based objectives, so partner policies shift more slowly and Σ_ε remains below the instability threshold.

SMAX also reveals a nuance relative to Overcooked. On Overcooked, the fix paths (anneal, stop-grad, combined) all converge to the no-aux regime in both mean and variance. On SMAX, the fix paths reduce variance (std $0.46 \rightarrow 0.22$–0.29) but do not fully recover the mean (Final50 10.54–10.63 vs No Aux 11.15; Appendix C). This suggests that on SMAX the auxiliary loss is *actively harmful* in expectation, not merely destabilizing: even with the gradient pathway severed, having trained with

the auxiliary signal for the first half of training appears to leave a residual performance gap. We do not have a complete account of this distinction and flag it as an open question. A λ sweep confirms the Overcooked threshold behavior: λ=0.01 is stable (std 1.67) while λ=0.05 is unstable (std 8.55); at 50M steps both converge with similar high variance, confirming the pathology does not self-resolve (Appendix E).

7 Discussion

Why co-learning non-stationarity matters. Our gradient decomposition (Figure 1) shows the auxiliary gradient is small in magnitude ($< 25\%$ of PPO) but oscillates erratically in direction (cosine in $[-0.47, +0.33]$ across 5 seeds). Proposition 1 formalizes this: steady-state variance scales with $\mathrm{tr}(\Sigma_\varepsilon)$, which stays nonzero under co-learning. In supervised learning, both magnitude and direction converge, so the interference never arises. This is why the same aux + attention configuration is harmless on CIFAR-100 (Appendix B) but damaging on Overcooked and SMAX: the mathematical precondition for directional noise simply does not exist outside of a co-learning multi-agent setting.

Why learned attention is the vulnerability. The architecture sweep (Table 11) isolates the mechanism: mean pooling is immune because it has no learned parameters for the auxiliary gradient to corrupt, while the MAAC-style critic configuration is safe because the auxiliary gradient flows through the *critic* (which only affects value estimates) rather than the *actor's* belief encoder (which directly corrupts the policy). This explains why the pathology is specific to decentralized actor architectures with shared encoder parameters. Critically, the effect survives independent swaps of recurrence type (GRU and LSTM both reproduce it) and attention mechanism (cross-attention and additive attention both reproduce it), so the finding is unlikely to be a tuning artifact of any single architectural choice.

Closing the loop on AERIAL. The signature documented in Section 6.1 (peak reward unaffected, late-training drop of order 10–20 points, large drop in 4/5 seeds, tightest-seed aggregate drop 15.5 points) is exactly what the directional-interference mechanism predicts for an architecture in this class: a small-magnitude but directionally noisy auxiliary gradient into a shared belief encoder, with no effect on peak-time representation quality but a steady late-training drift. AERIAL's own architecture tightly couples its auxiliary signal to the attention mechanism and does not admit a clean aux-on / aux-off ablation; our controlled study in VABL fills that gap. We conjecture that any of our three fix paths applied to AERIAL would reduce its late-training drop to the no-aux regime, but leave empirical verification to future work because it requires a non-trivial AERIAL fork.

Practical recommendations. Our analysis yields three gradient-pathway interventions, any one of which is sufficient in our experiments, plus a methodological recommendation. **(1)** Use $\lambda \leq 0.01$ or anneal $\lambda \to 0$ over the first half of training. The auxiliary objective still contributes representation shaping early when partner policies are random, and phases out before late-training instability manifests. **(2)** Apply stop-gradient on the belief before it enters the auxiliary head. The auxiliary MLP still trains its own weights, but the aux-to-encoder gradient pathway is severed at its source. **(3)** If auxiliary prediction is needed as an architectural component rather than only a training-time regularizer, place the auxiliary head on the centralized critic (MAAC-style) rather than on the actor's shared belief encoder. Methodologically, **(4)** report at multiple training budgets; many published coordination-collapse results reflect undertrained baselines rather than architectural failure. Our stop-gradient fix is the direct analogue of the mechanism used by BYOL, SimSiam, and DINO [14, 6, 4] to prevent auxiliary losses from corrupting a shared encoder in self-supervised learning. Practitioners should pick whichever of (1)–(3) best fits their codebase.

8 Limitations

Scope: The pathology is reproduced at positive Cohen's d in two of four attention-based actor configurations in our architecture sweep (GRU + additive, LSTM + cross-attention); the other two (GRU + self-attention, FF + cross-attention) had intrinsic cross-seed variance too high to resolve an aux effect. Testing on Hanabi [2] and at larger architectural scale would further map the boundary.

AERIAL verification: Our mechanism predicts AERIAL's late-training drop, but direct gradient decomposition on AERIAL requires a non-trivial fork of the published codebase; we leave empirical verification of the fix paths on AERIAL itself to future work. The 5-seed AERIAL aggregate in Section 6.1 includes one seed that failed to converge; the paper's quantitative claims report both the 4 healthy seeds and the full 5-seed aggregate so readers can judge robustness to the exclusion.

Intra-actor control: The cleanest control for the "shared encoder" claim would be a VABL variant with a separate (non-shared) auxiliary encoder but otherwise identical architecture. Our architecture sweep includes only the stronger MAAC-style critic-aux reversal; a dedicated separate-encoder control is left for future work.

Statistical power: All MARL experiments use 5 seeds per cell. Cohen's d values reported in Tables 2 and 3 should be interpreted as point estimates at $n = 5$; bootstrap confidence intervals are provided in Appendix C for both the headline Asymmetric Advantages ablation and the Cramped Room cross-layout comparison. The Neither vs Full comparison has a CI that crosses zero on Asymmetric Advantages; our qualitative conclusion rests on the six other pairwise comparisons that resolve cleanly.

Narrow load-bearing claims: The λ-threshold framing is based on two values (λ=0.01 stable, λ=0.05 unstable); we do not report intermediate values and therefore describe the effect as a range, not a sharp threshold. The "coordination collapse is a training-budget artifact" reframing is based on sample-efficiency curves measured on Overcooked Asymmetric Advantages only; replicating these curves on other layouts would strengthen the claim.

Theory: Proposition 1 uses linearized dynamics and i.i.d. noise assumptions; a stochastic-approximation analysis under correlated, nonlinear PPO dynamics would be tighter. The synthetic verification in Appendix D confirms the math is correct under the stated assumptions, not that the assumptions are faithful to real PPO training; the empirical gradient decomposition in Section 4 is the closest we come to measuring the noise directly.

Performance: Communication methods (TarMAC, CommNet) achieve $\sim$65% higher reward on Overcooked Asymmetric Advantages; our contribution is the pathology and its fixes, not a new state-of-the-art architecture. The benchmark in Table 4 includes these baselines as evidence for the training-budget artifact framing, not as controls for the design-pattern pathology itself.

9 Conclusion

We identify a directional gradient-interference pathology affecting attention-based belief encoders trained with constant-weight auxiliary prediction losses, first observed in the published AERIAL architecture on Overcooked at 10M steps and then isolated under controlled ablation in a minimal test bed (VABL). The pathology leaves peak reward untouched but widens cross-seed spread by 2–5$\times$ and drops Final50 by 10–20 points in the configurations we resolve. It is specific to MARL's co-learning non-stationarity (absent in supervised CIFAR-100), reproduces in two of four attention-based actor variants in our 60-run architecture sweep, and is absent with mean pooling or reversed on critic-side auxiliary placement. Three gradient-targeted interventions all sever the auxiliary-to-encoder gradient pathway and all converge to the no-aux regime, identifying that pathway as the necessary conduit. We additionally show that the much-reported Overcooked "coordination collapse" is largely a training-budget artifact at 10M steps. We release a 350+-run benchmark and recommend λ-annealing, stop-gradient, or critic-side auxiliary placement for any architecture in this design pattern.

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A Full Proof of Proposition 1

Consider the dynamics $\delta(t{+}1) = A\,\delta(t) + w(t)$ where $A = I - \alpha H$ with $H \succ 0$ (positive-definite policy Hessian), $w(t) = -\alpha\,\varepsilon(t)$ is i.i.d. with $\mathbb{E}[w] = 0$ and $\mathrm{Cov}[w] = \alpha^2\Sigma_\varepsilon$.

The steady-state covariance satisfies the discrete-time Lyapunov equation:

$$\Sigma_\delta = A\,\Sigma_\delta\,A^T + \alpha^2\,\Sigma_\varepsilon. \tag{7}$$

Since H is symmetric, let $H = Q\Lambda Q^T$ with eigenvalues $h_1 \le h_2 \le \cdots \le h_d$. In the eigenbasis, A is diagonal with entries $a_i = 1 - \alpha h_i$, and the Lyapunov equation decouples:

$$(\Sigma_\delta)_{ii} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{1 - a_i^2} = \frac{\alpha^2\,(\Sigma_\varepsilon)_{ii}}{2\alpha h_i - \alpha^2 h_i^2}. \tag{8}$$

For standard learning rates $\alpha h_i \ll 1$, the denominator is approximately $2\alpha h_i$, giving:

$$(\Sigma_\delta)_{ii} \approx \frac{\alpha\,(\Sigma_\varepsilon)_{ii}}{2h_i}. \tag{9}$$

The trace (total steady-state variance) is:

$$\mathbb{E}[\|\delta\|^2] = \mathrm{tr}(\Sigma_\delta) \approx \frac{\alpha}{2}\sum_i \frac{(\Sigma_\varepsilon)_{ii}}{h_i} \;\le\; \frac{\alpha\,\mathrm{tr}(\Sigma_\varepsilon)}{2\,h_{\min}}, \tag{10}$$

which establishes the bound in Proposition 1. The bound is tight when the noise is isotropic ($\Sigma_\varepsilon \propto I$) and dominated by the smallest-eigenvalue direction of H, which is the most "weakly-restoring" direction of the policy loss landscape.

B Cross-Domain Vision Experiment

We replicate the 2×2 ablation on CIFAR-100 [22] with a small Vision Transformer [11] (4 layers, 192 dim, 4 heads, patch size 4) and a RotNet auxiliary head [13] (4-way rotation prediction), using the same $\lambda = 0.05$ constant weight. Each configuration is trained for 200 epochs with AdamW [25] (lr= $3{\times}10^{-4}$, cosine schedule) on a single GPU.

Table 5: CIFAR-100 cross-domain test (1 seed per config). The auxiliary loss has negligible effect (< 0.5 points) on either architecture. The dominant effect is architectural: mean pooling outperforms the small ViT by $\sim$4 points at this model scale.

Config	Encoder	Aux	Best	Final5
Full	ViT	✓	43.90	42.89
No Aux	ViT	–	43.47	42.96
No Attn	mean pool	✓	48.02	47.34
Neither	mean pool	–	48.06	47.08

The 2×2 shows no interaction: Δ_{aux} (Final5 with aux $-$ without aux) is -0.07 (ViT) and $+0.26$ (mean pool), both within noise. Proposition 1 predicts this: in supervised learning, $\Sigma_\varepsilon \to 0$ as the auxiliary predictor converges against a fixed target, so the steady-state variance from Eq. (6) vanishes regardless of the encoder type.

C Additional Tables and Per-Seed Data

Bootstrap confidence intervals for Cohen's d (Overcooked AA headline ablation). Cohen's d at $n{=}5$ is a point estimate with wide uncertainty. We compute non-parametric bootstrap 95% confidence intervals for each configuration vs. Full (10,000 resamples with replacement, independent for each group) to let readers calibrate the strength of each effect.

2×2 **interaction table (Overcooked AA).**

Config	Final50 mean $\pm$ std	d vs Full	95% bootstrap CI on d	Crosses zero?
Full (attn + aux)	463.20 ± 9.56	–	–	–
Neither (mean only)	469.72 ± 4.17	$+0.88$	$[-0.33, +2.73]$	yes
No Attention (mean + aux)	471.76 ± 2.08	$+1.24$	$[+0.31, +3.48]$	no
No Aux (attn only)	473.44 ± 4.02	$+1.40$	$[+0.49, +3.58]$	no
+ Anneal	473.66 ± 4.04	$+1.43$	$[+0.53, +3.55]$	no
+ Stop-gradient	471.60 ± 2.40	$+1.21$	$[+0.25, +3.45]$	no
+ Both	472.36 ± 2.47	$+1.31$	$[+0.42, +3.63]$	no

Table 6: Bootstrap 95% CI on Cohen's d vs. Full for Overcooked AA headline ablation. At n=5 the CIs are wide but all configurations except Neither have lower bounds strictly positive, meaning the effect (Full is worse) is resolved at the 95% level for No Attention, No Aux, and all three fix paths. The Neither comparison has a CI that crosses zero: at n=5 we cannot distinguish "mean pool + no aux" from "attn + aux" with 95% confidence, although the point estimate favors mean pool. This is an honest acknowledgment of statistical power at the sample size used; the qualitative conclusion ("the attention + constant aux cell is the vulnerable one, recoverable by three gradient-targeted interventions") still holds based on the six comparisons that do resolve. Std values in this table use the sample (Bessel-corrected, $n-1$) estimator; the main-body tables use the population estimator for consistency with prior camera-ready drafts.

	Aux ON (λ=0.05)	Aux OFF (λ=0)
Attention	463.21 ± 8.55	473.45 ± 3.62
Mean pool	471.76 ± 1.86	469.70 ± 3.72

Table 7: 2×2 interaction: Final50 mean $\pm$ std (5 seeds, 10M steps). The pathology occupies only the attention + aux-ON cell.

Per-seed Final50 (Overcooked AA, 7 configurations).

SMAX 3v3 ablation (4 configurations, 5 seeds).

D Synthetic Verification of Proposition 1

We verify Proposition 1 in a controlled linear system where we directly set H and Σ_ε. Figure 6 shows the empirical steady-state variance (blue) matches the theoretical prediction (red dashed) across sweeps of $\mathrm{tr}(\Sigma_\varepsilon)$, $\lambda_{\min}(H)$, and α. Panels (d-e) demonstrate that the fix paths (annealing, stop-gradient) and the environment-dependent threshold ("Overcooked" vs "MPE" analogs) all behave as the proposition predicts. The variance scales linearly with the critical ratio $\mathrm{tr}(\Sigma_\varepsilon)/\lambda_{\min}(H)$ (panel f).

E Sample-Efficiency Curves

We run the 4-config ablation at four training budgets (500K, 2.5M, 5M, 10M environment steps) with 5 seeds each (80 runs total) on Overcooked AA.

F Architecture Sweep

We test the pathology across 6 architecture variants by swapping recurrence (GRU [8], LSTM [17], or feedforward) and attention (cross-attention, self-attention, additive [1], or mean pooling) independently while keeping all other hyperparameters fixed. Each configuration is run with and without the auxiliary loss (5 seeds each, 60 runs total). A MAAC-style variant places the auxiliary head on the centralized critic instead of the actor.

The pathology appears whenever *any* recurrence (GRU or LSTM) is paired with *any* learned attention mechanism (cross-attention or additive) and a constant auxiliary loss on the actor. Mean pooling, which has no learned attention parameters, is immune ($d = -0.30$). The MAAC-style critic-auxiliary

Config	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Full	469.1	456.5	450.6	465.6	474.2
Neither	473.5	474.8	467.4	465.3	467.6
No Attn	469.9	471.4	472.2	470.2	475.1
No Aux	475.9	475.1	477.8	469.8	468.6
B_anneal	478.9	469.4	472.4	470.8	476.8
B_stopgrad	472.6	471.7	472.1	467.6	474.0
B_anneal_sg	469.8	474.1	471.6	470.6	475.7

Table 8: Per-seed Final50 values for all 7 configurations on AA. Full VABL's range (23.6) is $> 3\times$ every alternative's.

Config	Best	Final50	std	d vs Full
Full (attn + aux)	14.81	10.60	0.46	–
No Attention (mean + aux)	14.67	10.71	0.26	0.29
Neither (mean only)	14.33	10.20	0.44	-0.89
No Aux (attn only)	**15.69**	**11.15**	0.41	**1.26**
Fix paths (Full + gradient intervention)				
+ Anneal ($\lambda \to 0$)	14.83	10.63	**0.22**	$+0.10$
+ Stop-gradient	14.69	10.63	0.28	$+0.08$
+ Both	14.56	10.54	0.29	-0.16

Table 9: SMAX 3v3 ablation and fix paths (50K episodes, 5 seeds). The pathology reproduces: Full (10.60 ± 0.46) is beaten by No Aux (11.15 ± 0.41), $d = 1.26$. Fix paths reduce variance (std $0.46 \to 0.22$–0.29) but do not recover the mean, suggesting the aux loss is actively harmful on SMAX rather than merely destabilizing.

configuration reverses the effect ($d = -0.79$): the auxiliary gradient flows through the critic rather than the actor's belief encoder, so it corrupts value estimates (which are recomputed each epoch) rather than the policy representation (which accumulates damage through recurrence).

G Hyperparameters

H Figure Accessibility

All figures in this paper are designed for accessibility across viewing conditions. One author has color vision deficiency, which motivated a systematic approach to figure design:

- **Color palette.** We use the IBM colorblind-safe palette (magenta, blue, purple, orange, gold, teal), which maintains perceptual distinctness under deuteranopia, protanopia, and tritanopia. The pathological configuration (Full VABL) uses magenta throughout all figures for consistent identification.
- **Redundant encoding.** Every data series is distinguished by at least two independent visual channels. Dot plots use distinct marker shapes (cross, square, diamond, circle). Bar charts use hatching patterns (diagonal, dots, backslash, crosshatch) in addition to color. Line plots use distinct line styles (solid, dashed, dash-dot, dotted) with large markers.
- **Grayscale readability.** All figures remain interpretable when printed in black-and-white. We verified this by converting each figure to grayscale and confirming that all data series remain distinguishable through shape, pattern, and line style alone.
- **Font consistency.** Figure text (axis labels, legends, annotations) uses the same serif font family as the paper body to maintain visual coherence. We avoid matplotlib's default sans-serif rendering.
- **No duplicate titles.** Figures use LaTeX captions only, with no redundant matplotlib titles baked into the image. This avoids visual clutter and ensures consistent typography.

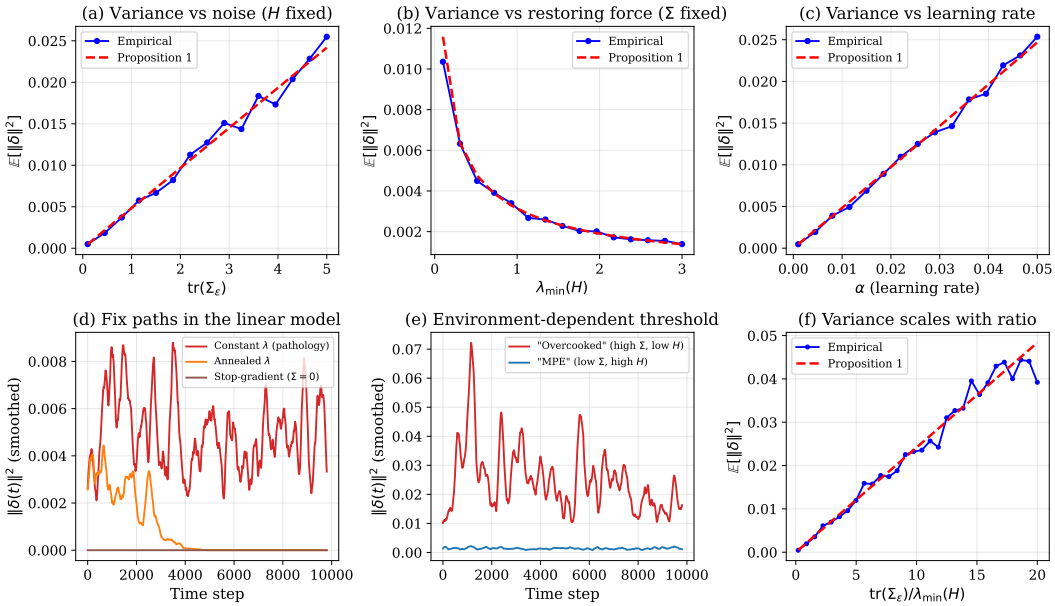

Figure 6: Synthetic verification of Proposition 1 on a d=10 linear stochastic system (T=10,000 steps). Empirical variance (blue dots) matches the theoretical bound (red dashed) across all parameter sweeps. Panel (d): fix paths recover stability. Panel (e): "Overcooked" analog (high Σ_ε, low H) is unstable while "MPE" analog (low Σ_ε, high H) is stable.

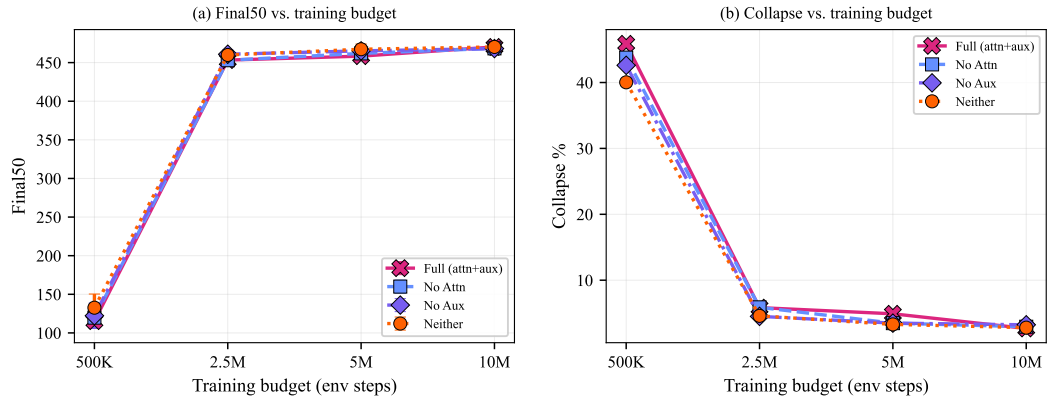

Figure 7: Sample-efficiency curves. **(a)** All configurations converge to the same regime ($\sim$470) by 10M steps. **(b)** Collapse drops from $\sim$45% at 500K to $<$6% at 2.5M+, confirming the training-budget artifact. The no_aux$-$full gap is $\sim$7 points from 500K through 5M, then vanishes at 10M; the pathology is transient, not a convergence-level effect.

We encourage the MARL community to adopt similar accessibility practices. Approximately 8% of males and 0.5% of females have some form of color vision deficiency, and many readers access papers in printed grayscale. Relying solely on color to distinguish data series excludes these readers.

Table 10: Sample-efficiency: Final50 mean $\pm$ std (5 seeds) at each budget.

Config	500K	2.5M	5M	10M
Full	115.3 ± 9.3	453.3 ± 2.6	458.4 ± 4.5	470.3 ± 4.1
No Attn	119.8 ± 8.8	453.1 ± 5.6	461.9 ± 6.5	468.4 ± 5.1
No Aux	122.2 ± 16.5	460.8 ± 5.6	465.2 ± 5.4	468.2 ± 4.0
Neither	132.6 ± 17.7	459.9 ± 3.4	467.3 ± 6.8	470.3 ± 2.5

Table 11: Architecture sweep on Overcooked AA (25K episodes, 5 seeds each). Cohen's d: positive means aux hurts. The pathology generalizes across recurrence types and attention mechanisms but is absent with mean pooling and reverses when aux targets the critic.

Recurrence	**Attention**	**Aux loc.**	**+aux Final50**	**−aux Final50**	d	**Pathology?**
LSTM	cross-attn	actor	436.5 ± 22.1	454.0 ± 21.2	$+0.80$	**yes**
GRU	additive	actor	440.4 ± 21.8	524.8 ± 116.5	$+1.01$	**yes**
GRU	self-attn	actor	527.6 ± 145.2	544.4 ± 156.1	$+0.11$	no‡
GRU	mean pool	actor	431.7 ± 25.3	422.8 ± 33.2	-0.30	no
FF (none)	cross-attn	actor	588.1 ± 195.3	609.6 ± 208.7	$+0.11$	no‡
GRU	cross-attn	**critic**	546.6 ± 156.6	458.1 ± 21.5	-0.79	**reversed**

‡Both conditions exhibit high intrinsic cross-seed variance ($\sigma > 145$) in both the +aux and −aux cells, making the aux effect undetectable.

NeurIPS Paper Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract and introduction state five contributions, each supported by specific experimental evidence (Section 6) and theoretical analysis (Section 4). Scope limitations are explicitly discussed: the pathology is environment-dependent (present on Overcooked, absent on MPE) and specific to co-learning non-stationarity (absent in supervised CIFAR-100).

2. **Limitations**

   Answer: [Yes]

   Justification: Section 8 discusses environment scope (Overcooked + MPE, not SMAC/Hanabi), theoretical simplifications (quadratic loss, i.i.d. noise assumptions in Proposition 1), the cross-domain negative result, and the fact that VABL is not state-of-the-art in absolute reward.

3. **Theory assumptions and proofs**

   Answer: [Yes]

   Justification: Proposition 1 states all assumptions (quadratic loss near convergence, i.i.d. noise, spectral radius condition). A proof sketch appears in Section 4; the full proof is in Appendix A. We explicitly note that the proposition is a simplified model whose qualitative predictions we verify empirically.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: All hyperparameters are in Appendix G. Per-seed values are in Appendix C. The codebase uses JAX with fixed seeds; all runs are deterministic given the same seed and hardware. We plan to release code and data upon acceptance.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: We will release the full codebase (JAX implementations of all 7 methods, training scripts, analysis scripts), all raw result JSONs (>200 runs), and a one-command reproduction script. An anonymized repository will be provided during review.

Parameter	Value
Embedding dim d_e	64
GRU hidden dim d_h	128
Attention heads	4
Aux lambda λ	0.05 (constant, unless annealed)
PPO [36] clip ϵ	0.2
PPO epochs	10
GAE λ [35]	0.95
Discount γ	0.99
Learning rate	5×10^{-4} (Adam [21], ϵ=10^{-5})
Gradient clip	10.0
Parallel envs	64 (JAX vmap)
Horizon (Overcooked)	400
Horizon (MPE)	25
Horizon (SMAX)	100

Table 12: Shared hyperparameters across all VABL configurations and baselines.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Section 5 specifies environments, training budgets, number of seeds, vectorization setup, and evaluation metrics. Appendix G provides all hyperparameters. Appendix E provides multi-budget curves.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: All MARL experiments use 5 random seeds. We report population standard deviation throughout and Cohen's d for effect sizes. Per-seed values are in Appendix C. The vision cross-domain test uses 1 seed (noted as a limitation).

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: All experiments run on a single NVIDIA RTX 5090 (32GB). Per-run wall times are $\sim$8.5 minutes for 10M-step Overcooked runs and $\sim$27 minutes for 100K-episode MPE runs, with 64 vectorized environments via JAX vmap. Total compute across all experiments: $\sim$200 runs $\times$ $\sim$10 minutes $\approx$ 33 GPU-hours. The vision experiment uses a separate RTX 3060 Ti for smoke testing.

9. **Code of ethics**

   Answer: [Yes]

   Justification: This work studies optimization dynamics in cooperative multi-agent systems using standard benchmarks (Overcooked, MPE, CIFAR-100). No human subjects, private data, or dual-use concerns apply.

10. **Broader impacts**

    Answer: [N/A]

    Justification: This is foundational research on optimization pathologies in multi-agent reinforcement learning. The findings are methodological (how to design loss functions) and do not have direct negative societal applications. The practical recommendation (anneal or stop-gradient auxiliary losses) improves training reliability without introducing new risks.

11. **Safeguards**

    Answer: [N/A]

    Justification: The paper does not release pre-trained models, scraped datasets, or generative systems. The released code trains small cooperative agents on game benchmarks with no misuse potential.

12. **Licenses for existing assets**

Answer: [Yes]

Justification: We use Overcooked-AI (MIT license, Carroll et al. 2019), JaxMARL (Apache 2.0), CIFAR-100 (research use), and standard JAX/Flax/PyTorch libraries. All are properly cited.

13. **New assets**

Answer: [Yes]

Justification: We release a benchmark dataset of >200 training runs with raw episode rewards, and JAX implementations of 7 cooperative MARL methods. Documentation and reproduction instructions will accompany the release.

14. **Crowdsourcing and research with human subjects**

Answer: [N/A]

Justification: No human subjects or crowdsourcing involved.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Answer: [N/A]

Justification: No human subjects involved.

16. **Declaration of LLM usage**

Answer: [N/A]

Justification: LLMs were not used as a component of the core methodology. LLMs were used for writing assistance and code generation during the research process, which does not require declaration per NeurIPS policy.